RECOMPLETION IS COMPATIBLE WITH BASE CHANGE OF FORMAL ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The recompletion of a formal algebraic space along a closed subset of its reduced space. Same chapter, the construction preceding the lemma (proof line 8061, 'by construction'; source lines 7960-8044). Supplies the objects X'_{/T'} and X_{/T} and the fact that the horizontal arrows are monomorphisms by construction.
- Formal algebraic spaces over S and their morphisms as sheaves on (Sch/S)_fppf, and the reduced space of a formal algebraic space. Same chapter, Definition definition-formal-algebraic-space and the discussion of the reduced space preceding it. Lets the cartesian property be tested on morphisms from schemes into X' and gives the meaning of |X_red| and of f_red.

Lemma 0.1. *Let S be a scheme. Let $f : X' \to X$ be a morphism of formal algebraic spaces over S. Let $T \subset |X_{red}|$ be a closed subset and let $T' = |f_{red}|^{-1}(T) \subset |X'_{red}|$. Then*

$$\begin{array}{ccc} X'_{/T'} & \longrightarrow & X' \\ \downarrow & & \downarrow f \\ X_{/T} & \longrightarrow & X \end{array}$$

is a cartesian diagram of formal algebraic spaces over S.

Proof. Namely, observe that the horizontal arrows are monomorphisms by construction. Thus it suffices to show that a morphism $g : U \to X'$ from a scheme U defines a point of $X'_{/T}$ if and only if $f \circ g$ defines a point of $X_{/T}$. In other words, we have to show that $g(U)$ is contained in $T' \subset |X'_{red}|$ if and only if $(f \circ g)(U)$ is contained in $T \subset |X_{red}|$. This follows immediately from our choice of T' as the inverse image of T. □

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